

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/377121829>

Comparative genomics and characterization of a multidrug-resistant *Acinetobacter baumannii* VRL-M19 isolated from a crowded setting in India

Article in *Infection Genetics and Evolution* · January 2024

DOI: 10.1016/j.meegid.2023.105549

CITATIONS

3

READS

49

4 authors, including:



Tina Sharma

Centre for Development of Advanced Computing

10 PUBLICATIONS 16 CITATIONS

[SEE PROFILE](#)



Anshu Bhardwaj

Open Source Drug Discovery Unit, Council of Scientific and Industrial Research (CSIR...)

98 PUBLICATIONS 908 CITATIONS

[SEE PROFILE](#)



Vemuluri Venkata Ramana

CSIR-Institute of Microbial Technology

55 PUBLICATIONS 763 CITATIONS

[SEE PROFILE](#)

Journal Pre-proof

Comparative genomics and characterization of a multidrug-resistant *Acinetobacter baumannii* VRL-M19 isolated from a crowded setting in India

Neha Verma, Tina Sharma, Anshu Bhardwaj, Vemuluri Venkata Ramana



PII: S1567-1348(23)00147-8

DOI: <https://doi.org/10.1016/j.meegid.2023.105549>

Reference: MEEGID 105549

To appear in: *Infection, Genetics and Evolution*

Received date: 31 July 2023

Revised date: 17 December 2023

Accepted date: 31 December 2023

Please cite this article as: N. Verma, T. Sharma, A. Bhardwaj, et al., Comparative genomics and characterization of a multidrug-resistant *Acinetobacter baumannii* VRL-M19 isolated from a crowded setting in India, *Infection, Genetics and Evolution* (2023), <https://doi.org/10.1016/j.meegid.2023.105549>

This is a PDF file of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability, but it is not yet the definitive version of record. This version will undergo additional copyediting, typesetting and review before it is published in its final form, but we are providing this version to give early visibility of the article. Please note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

Comparative genomics and characterization of a multidrug-resistant *Acinetobacter baumannii* VRL-M19 isolated from a crowded setting in India

^{1,3}Neha Verma, ^{2,3}Tina Sharma, ^{2,3}Anshu Bhardwaj*, ^{1,3}Vemuluri Venkata Ramana*

¹Microbial Type Culture Collection and Gene Bank (MTCC), CSIR-Institute of Microbial Technology, Chandigarh-160036, India.

²Bioinformatics Centre, CSIR-Institute of Microbial Technology, Chandigarh-160036, India.

³Academy of Scientific and Innovative Research (AcSIR), Ghaziabad-201002, India.

*** Corresponding authors**

venkat@imtech.res.in

+91-1722880-749, +91-9545067768

anshu@imtech.res.in

Abstract

A crowded vegetable market serves as a mass gathering, posing a potential risk for infection transmission. In this study, we isolated a multidrug-resistant *Acinetobacter baumannii* strain, VRL-M19, from the air of such a market and conducted comparative genomics and phenotypic characterization. Antimicrobial susceptibility testing, genome sequencing using Illumina HiSeq X10, and pan-genome analysis with 788 clinical isolates identified core, accessory, and unique drug-resistant determinants. Mutational analysis of drug-resistance genes, virulence factor annotation, in vitro pathogenicity assessment, subsystem analysis, Multilocus sequence typing, and whole genome phylogenetic analysis were performed. VRL-M19 exhibited multidrug resistance with 69 determinants, and analysis across 788 clinical isolates and 350 Indian isolates revealed more accessory genes (52 out of 6%) in the Indian isolates. Multiple mutations were observed in drug target modification genes, and the strain was identified as a moderate biofilm-former with 55 virulence factors. Whole genome phylogenetics indicated a close relationship between VRL-M19 and clinical *A. baumannii* strains. In conclusion, our comprehensive study suggests that VRL-M19 is a multidrug-resistant, potential pathogen with biofilm-forming capabilities, closely associated with clinical *A. baumannii* strains.

Keywords

Acinetobacter baumannii, antimicrobial resistance, virulence factors, mass gathering, Pangenome

1. Introduction

Antimicrobial resistance (AMR), a rising global concern and a multifarious problem. One health approach for the containment of AMR required an expanded approach. Whereas in India, the rate of AMR increases in humans, animals, and the environment disproportionately in the past few years. Such a scenario poses a challenge for the containment of AMR and makes India ‘the AMR capital of the world’(Taneja and Sharma, 2019). Multidrug resistance (MDR) is persistently expanding with time in diverse ecosystems. Most of such observations are made in clinical isolates followed by environmental strains. It is now well acknowledged that in order to understand the evolutionary route from environmental to opportunistic pathogen lifestyle, as seen in *Acinetobacter baumannii* (*A. baumannii*), it is imperative to characterize *A. baumannii* in natural reservoirs. There is still a considerable gap in understanding the dispersion of AMR genes among populations and the dynamics of outbreaks in non-clinical settings, such as the mass gathering of the human population. According to WHO, a mass gathering is defined as an occasion of a large number of people either organized or spontaneous events in a confined space and time (Memish et al., 2013). Several reports suggest that mass gathering is one of the social activities provoking the transmission of pathogens among the population in a defined environment (Gautret and Steffen, 2016; Leangapichart et al., 2016; Wharton et al., 1990). There are various reports of clinical isolates of *A. baumannii* across different regions in India. A study in Sitapura, Jaipur, Rajasthan, reveals a high prevalence of *A. baumannii* strains, with a majority exhibiting resistance to multiple antibiotics, classifying it as an emerging hospital-acquired pathogen (Sharma et al., 2018). Another study in Varanasi highlights the increasing emergence of *A. baumannii* in hospital-acquired pneumonia, with a significant proportion showing

resistance to imipenem (Banerjee et al., 2018). Limited studies in the Punjab region also indicate antibiotic resistance, with *A. baumannii* isolates from Chandigarh displaying resistance to meropenem and imipenem (Vijayakumar et al., 2019). These studies underscore the diverse geographical distribution and antibiotic resistance patterns of *A. baumannii* infections in different regions of India.

In the current study, an *A. baumannii* strain VRL-M19 was isolated from the air of a crowded fruit and vegetable market where more than 1200 persons gathered at a time, which could be a favorable niche for the transfer of aerosols (containing microorganisms from the oral cavities and respiratory tracts), among the attendees, during involuntary activities such as coughing, sneezing, respiratory activities and etc. Such activities enhance the chance of transmission of human visceral microbiota among individuals as well as into the environment (da Silva Nali et al., 2016; Sebastiaan J van Hest et al., 2009). *A. baumannii* is an opportunistic pathogen known to cause various lethal infections. It shows tolerance to wide ranges of pH, humidity, and temperature, and is also able to survive on dry surfaces for several months which can cause serious outbreak infections (Almaghrabi et al., 2018). *A. baumannii* emerged as a significant multidrug-resistant pathogen worldwide and its resistance increasingly reported during the last decade. Resistant *A. baumannii* is an emerging challenge due to its resistance patterns to most of the antibiotics widely used in the treatment of *A. baumannii* infection (Singh et al., 2013). Available literature reported the resistance profile of *A. baumannii* against a significant number of antibiotics (Badave and Dhananjay, 2015; Singh et al., 2013; Vrancianu et al., 2020). Though several studies have indicated an upward trend in the prevalence of MDR *A. baumannii*, the evolutionary rate of resistance varies widely depending on its habitat (Yadav et al., 2020).

A. baumannii, a nosocomial, is opportunistic among ESKAPE pathogens (*Enterococcus faecium*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, and *Enterobacter* species), which acquired significant clinical attention due to its rapid increase in multidrug resistance (Santajit and Indrawattana, 2016). Whereas, the studies about its occurrence as a potential reservoir in environmental and non-clinical settings are obscure. Horizontal gene transfer of drug-resistant determinants among the environmental and clinical strains may be one of the sources for the persistent genetic modifications in the genome of *A. baumannii*, which is prone to acquiring drug resistance. Hence, it is important to study the MDR *A. baumannii* strain isolated from diverse environments to understand its drug-resistant profiles and pathogenesis. In view of this, in the current study, we report the genetic characterization and comparative assessment of MDR *A. baumannii* VRL-M19 isolated from an open-air environment. Comparative genomic analysis of VRL-M19 with 788 strains of *A. baumannii* reveals the presence of several virulence factors, drug resistance, and mutations among the genes associated with the target modification resistance mechanism.

2. Material and methods

2.1 Isolation and identification

Strain VRL-M19 was isolated from air by the settle plate method at an open-air vegetable and fruit market of sector 40 A, Chandigarh, India (30°73'93" N, 76°73'24" E). For air sampling, various sterile media in Petri plates (such as MacConkey agar, Reasoner's 2A agar (R2A), actinomycetes isolation agar, yeast mannitol agar, Sabouraud dextrose agar, tryptone soy agar, cetrimide agar base, eosin methylene blue agar, Lewenstein Jensen agar, gelatin agar, brain heart infusion agar, soil extract agar, and potato dextrose agar, *Streptomyces* agar) were exposed to the open area of the market for 3-4 h and then plates were incubated for 24-72 h at 37°C in the

laboratory. Various bacterial colonies grown on plates were picked on the basis of morphological variation and further purified on the respective media. Based on the antimicrobial resistance pattern of isolated strains, strain VRL-M19 has been selected for further characterization. Strain VRL-M19 was identified through 16S rRNA gene sequencing by using a pair of universal bacterial primers i. e. 27F' (5'-AGAGTTTGATCMTGGCTCAG-3') and 1492R' (5'-ACGGCTACCTTGTTACGACT-3') (Weisburg et al., 1991; Yoon et al., 2017).

2.2. Antibiotic susceptibility testing

The antibiotic susceptibility test (AST) of VRL-M19 was determined using the disk diffusion method, conducted on Mueller-Hinton agar plates (Flemer, 2019). The antibiogram was executed against a panel of 37 antibiotics belonging to 12 classes, including four antibiotic-adjuvant combinations as specified in Table 1. The results were interpreted based on the zone of inhibition as per the guidelines available in Clinical Laboratory and Standard Institute (Weinstein and Clinical and Laboratory Standards Institute, n.d.).

2.3 Genome sequencing, quality assessment, and annotation

For genomic characterization, whole genome sequencing was performed using Illumina HiSeq X10 sequencing platform via paired-end 2x150 base pair (bp) chemistry and the resultant reads were assembled by Unicycler assembler with default k-mer sizes. The genome sequence of *A. baumannii* ATCC 17978-mff (NZ_CP012004.1) was used as a reference strain for comparative analysis in the current study (Zhou et al., 2021). Contig re-ordering and alignment were done using Mauve Contig Mover (MCM) followed by a quality check by QUAST 5.0.2. The parameter--strict -NA of QUAST 5.0.2 was used to compute NAX and NGAX statistics

(Mikheenko et al., 2018; Rissman et al., 2009). As per EUCAST (European Committee for Antimicrobial Susceptibility Testing) criteria, genome fraction of $\geq 75\%$, NGA50 ≥ 20000 , and the number of contigs ≤ 500 was used to assess the quality of VRL-M19 draft genome sequence (Ellington et al., 2017; Rissman et al., 2009). The resulting genome was annotated by using the RASTtk and the subsystem mapping was done using the online RAST server (Brettin et al., 2015).

2.3.1 Identification of IS element

The genome of VRL-M19 is scanned for insertion sequence (IS) elements using the ISEscan tool which provides the ORF coordinates for complete, as well as partial IS elements across the genome sequence and *in-house* algorithm in PFRL (https://github.com/tinabioinfo/VRL-M19/blob/main/Code_to_cover_DR_gene_with_IS_elements) was designed to map the genomic location of IS elements with respect to the Drug resistance determinants (Xie and Tang, 2017).

2.3.2 MLST and phylogenetic tree construction

Multilocus sequence typing (MLST) of the VRL-M19 was done via the online MLST (2.0.4) tool of the Center for Genomic Epidemiology (<https://cge.cbs.dtu.dk/services/MLST/>) using both Pasteur and Oxford schemes. The server used the database of PubMLST.org (Larsen et al., 2012). For genome sequence-based phylogenetic tree construction, 53 genomes of *A. baumannii* were downloaded from the NCBI (based on 16S gene similarity), and the genome of the Sequence type (ST624) strain of VRL-M19 was retrieved from the PubMLST database. A total of 55 genomes of *A. baumannii* including VRL-M19 were used to create a dendrogram by using

PATRIC online server (3.6.12) with default parameters except for a customized parameter, i.e., 1000 genes were chosen for the phylogeny. The codon tree uses amino acid and coding DNA of PATRIC (PGFams) protein family across the chosen genome group and aligned the protein sequences using MUSCLE, and the coding DNA with the codon align function of bioPython. Finally, the aligned sequences were analyzed by the program RAxML (Maximum likelihood-based phylogeny) using 100 rounds of “bootstrapping” (Antonopoulos et al., 2019).

2.3.3 Identification of virulence factors

The VFDB database and PATRIC were used to identify the virulence factors of strain VRL-M19 (Antonopoulos et al., 2019; Chen et al., 2016). From the VFDB database, the protein sequences of virulence factors for *A. baumannii* were retrieved and aligned with the CDSs of VRL-M19 with $\geq 50\%$ percentage identity, $\geq 90\%$ query coverage and 0.0001 e-value using BLASTp. The operon prediction was done using the fgenesB tool

(<http://linux1.softberry.com/berry.nhtml?topic=fgenesb&group=programs&subgroup=gfindb>)

with 11 genetic code and the closest organism ‘bacterial generic’ and all the predicted virulence factor genes were mapped to the operons of VRL-M19. The order of the virulence factors was cross-checked with the operons of the reference strain *A. baumannii* ATCC 17978-mff (**NZ_CP012004.1**) predicted using the fgenesB tool. BLAST Ring Image Generator (BRIG) was used to create a circular image for the visualization of the order and position of virulence factors over the VRL-M19 genome (Alikhan et al., 2011). Furthermore, the whole genome of VRL-M19 has also been checked for the presence of pathogenic protein families by using an online Pathogen Finder (version 1.1) tool (<https://cge.food.dtu.dk/services/PathogenFinder/>) to assess its pathogenesis in humans (Cosentino et al., 2013).

2.3.4 Identification of drug resistance genes

Identification of drug-resistance (DR) determinants was done using PATRIC, CARD, ResFinder, PRAP, and ABRicate (Antonopoulos et al., 2019; Bortolaia et al., 2020; Feldgarden et al., 2019; He et al., 2020; Jia et al., 2017). The PRAP tool was used to scan the VRL-M19 genome against CARD and Resfinder with 50% percentage identity and $\geq 90\%$ query coverage for protein sequences of VRL-M19. The ABRicate tool was used to scan the VRL-M19 contigs against CARD and NCBI databases.

2.4 Pangenome analysis of drug resistance genes

For the pangenome analysis of drug resistance genes in *Acinetobacter baumannii*, a comprehensive dataset of 788 high-quality genome sequences was assembled, featuring clinical isolates renowned for their pathogenicity in humans and documented MIC values following CLSI standards. The selection process incorporated EUCAST metrics to ensure sequences with CheckM contamination below 5% and CheckM completeness above 90%. These genomes were sourced from the NCBI and PATRIC databases, as detailed in previous studies (Antonopoulos et al., 2019; Sayers et al., 2022). Additionally, a curated set of 350 high-quality genomes from India, encompassing 39 complete genomes and 311 WGS sequences, was scrutinized. The resistance profiles of these Indian isolates were authenticated using computational methods (AdaBoost) and cataloged in the PATRIC database. Both datasets, along with VRL-M19, were subjected to pangenome analysis using the BPGA (Bacterial Pan Genome Analysis tool) pipeline, applying default parameters as described in earlier works (Chaudhari et al., 2016; Sharma et al., n.d.).

2.4.1 Mutational and domain analysis of core drug resistance genes

The core target modifications drug resistance genes obtained from pangenome analysis from all the selected clinical isolates of *A. baumannii* and VRL-M19 were aligned using ClustalW (Thompson et al., 1994). The resulting fasta format alignment file was subjected to variation analysis using an snp-sites tool which generated a vcf file with mutations mapped on VRL-M19 based on the reference genome (Page et al., 2016). Furthermore, domain analysis was performed using the Pfam 34.0 database with the E value of 1.0 to evaluate the impact of point mutations found in drug resistance genes that are associated with the target modification mechanism (Finn et al., 2014).

2.5 *In-Vitro* pathogenesis assays

To evaluate the phenotypic pathogenicity of the VRL-M19 strain, *in-vitro* assays were performed.

2.5.1 Biofilm formation

Ability of *A. baumannii* to adhere to abiotic surfaces to form biofilm is one of the important factors that supports its pathogenesis (Eijkelkamp et al., 2011; Gedefie et al., 2021). Biofilm formation of *A. baumannii* VRL-M19 was analyzed in a microtiter plate biofilm assay (Zeighami et al., 2019). Briefly, the culture was grown overnight in tryptic soy broth with 0.25% of glucose at 37°C. The free cells were removed and the biofilms were washed thrice with Phosphate buffer saline (PBS) and fixed with 99% (V/V) methanol. The wells were further stained with 1% (w/v) of crystal violet at room temperature for 20 minutes. The Crystal violet was dissolved using ethanol/acetone (80,20, v/v) for 20 minutes and the absorbance was measured at 595 nm. The assay was done in technical triplicates and biological duplicates of the VRL-M19 culture. The biofilm formation was measured as per the optical density (O.D) as follows: non-biofilm former

(<1.0), weak biofilm former (1.0 - 2.0), moderate biofilm former (2.0 - 3.0), strong biofilm former (>3.0).

2.5.2 Motility

Motility is often associated with bacterial virulence in various bacterial strains (Han et al., 2008; Alarcon et al., 2009) and also reported in several clinical isolates of *A.baumannii* (Eijkelkamp et al., 2011). Motility was analyzed by swimming agar plate method. For this, LB media and 0.3% agar plate were prepared. A bacterial culture of VRL-M19, which was grown overnight, was used for the motility analysis. Using a sterile micropipette, 2 μ L of the VRL-M19 culture was carefully placed in the center of the LB agar plate. This inoculation at the center allows for the observation and measurement of bacterial motility as the bacteria move outward from the point of inoculation. The plates were then incubated at 37 °C for 24 hours. To validate the motility assay, a nonmotile culture of *Staphylococcus aureus* MTCC 1430 was used as a negative control. The plates were examined for the presence and pattern of bacterial motility. Motile bacteria typically exhibit outward migration from the point of inoculation, creating a visible "swimming" pattern in the agar (Harding et al., 2013).

3. Results and discussion

In this study, we report a unique strain of *A. baumannii*, that is collected from air sampling, named VRL-M19. Both Antibiotic susceptibility test and genomics characterization of this strain is performed to map the drug resistance profile as well its phenotypic resistance. In order to gain further insights into the strain genotype, its virulence gene analysis, MLST and phylogeny affiliations was analyzed. Further the pangenome analysis is performed with around 788 clinical isolates of *A. baumannii*. Observations from various characterization techniques are reported

below.

3.1. Isolation and identification of VRL-M19

Strain VRL-M19 was isolated from air sample collected at a crowded fruit and vegetable market, Sector 40, Chandigarh, India. The 16S rRNA gene sequence similarity and whole genome DNA-DNA hybridization of VRL-M19 with *A. baumannii* ATCC 19606^T showed 99.9 % and 99.3 % respectively, which confirmed the taxonomic affiliation of VRL-M19 to *A. baumannii*. Further, the antibiotyping of VRL-M19 is studied extensively and explored the genotype-phenotype correlations. The genome sequence and 16S rRNA gene sequences of *A. baumannii* VRL-M19 have been deposited to GenBank under accession no. MHLEI000000000.1 and OQ692086 respectively.

3.2. Antibiotic resistance profile of VRL-M19

Strain VRL-M19 showed phenotypic resistance to eight antibiotics belonging to seven different classes such as beta-lactam, sulphonamide, macrolide, fluoroquinolones, nitrofurantoin, monobactam, and cephalosporin (**Table 1**). While it was susceptible to 29 antibiotics (including 4 antibiotic-adjuvant combinations) among a total of 37 antibiotics that are used for AST in this study (**Table 1**). VRL-M19, isolated from an open-air vegetable and fruit market during a mass gathering was found to be a MDR *A. baumannii* based on its resistance pattern.

3.3. Genome assembly and annotation of VRL-M19

The whole genome was assembled into a scaffold of 39 30 429 bp length, with a G+C content of 38.93 %. The QUAST analysis resulted in 18 contigs with a length of at least ≥ 1000 bp for

each contig. The largest contig was found to be a length of 63 6544 bp. The genome quality was determined by metrics of NGA50(=31917), and Genome fraction (85.879 %), which fulfilled the criteria of a good-quality genome as recommended by EUCAST. Global alignments were visualized via MCM as locally collinear blocks (LCBs), which represent the regions with homology without any rearrangement between genomes. LCBs are colored according to the homology of the reference genome (**Figure 1**). A total of 19 LCBs that correspond to the homologous regions that are shared by the two genome sequences were observed.

3.3.1 IS elements in VRL-M19 genome

ISEscan analysis of the concatenated and re-ordered contigs of strain VRL-M19 has shown four insertion sequence (IS) elements corresponding to three families: IS481 (IS481_30|IS481|protein: vir:99858 (start position:1421755, end position:1423512)), IS481 (IS481_213 (start position:2775756, end position:2776921)), IS21 (IS21_259 (start position:3891202, end position:3892965)), IS3 (IS3_61 (startposition:3928991, end position:3930373)). An *in-house* Perl algorithm did not identify any of these IS elements occurring at 1000 bp upstream of the identified drug resistance determinants. This observation indicates the absence of IS element nearby any of these drug resistance genes of VRL-M19.

3.3.2 Sequence Typing and Phylogeny of VRL-M19

MLST server assigns VRL-M19 to Sequence Type (ST) 624 based on only the Pasteur scheme. **Supplementary Table 1 and 2**). Two other strains, IOMTU501 and G20251192, of *A. baumannii* with ST624 were found in the PubMLST database, both of which were isolated from geographically distant origins while belonging to the same Sequence type. The whole genome

phylogenetic tree revealed the closest cladding of VRL-M19 with various *A. baumannii* strains of clinical origin. Strain VRL-M19 shows the closest relationship with an *A. baumannii* strain isolated from the lower respiratory tract of a human being. Thus, MLST and phylogeny relation predicts its origin from a clinical setup. **(Figure 2)**

3.3.3 Subsystem mapping

The RASTtk Pipeline was used for genome annotation and it predicted 3672 coding sequences (CDSs) and 69 RNAs. The RAST SEED server grouped only 44 % of the protein-coding sequences (1640 genes out of 3672) into a functional subsystem in the VRL-M19 genome. These 1640 protein-coding sequences were mapped to 309 subsystems, of which amino acids and derivatives ($n = 281$ CDSs), protein metabolism ($n = 187$), carbohydrates ($n = 177$), cofactors, vitamins, prosthetic groups, and pigments ($n = 140$), metabolism of aromatic compounds ($n=98$), membrane transport ($n = 86$), respiration ($n = 82$), fatty acids, lipids, and isoprenoids ($n = 78$), nucleosides and nucleotides ($n = 71$), stress response ($n = 64$), DNA metabolism ($n=56$), RNA metabolism ($n = 50$), sulfur metabolism ($n=48$), Miscellaneous ($n=43$), virulence, disease and defense ($n=33$), regulation and cell signaling ($n=32$), secondary metabolites ($n=23$), DNA metabolism ($n=56$), etc. have been identified. The list of cellular processes and the distribution of protein-coding genes are shown in **Figure 3**.

The maximum length of the protein sequence annotated is 5181 amino acids long (**WP_216116445.1**), which was predicted as a large exoprotein involved in heme utilization or adhesion. The large exoprotein is reported as a part of the Two partner system (TPS) family. TPS systems are known to be found among Gram-negative bacteria and are part of the Type Vb secretion system which is composed of two proteins, TpsB (AbFhaB), the outer membrane pore-

forming transporter, and TpsA (AbFhaC), the cognate secreted exoprotein that is involved in adhesion and biofilm formation. This TPS system was also reported in one of the highly adhesive known clinical strains AbH12O-A2 of *A. baumannii* responsible for a large outbreak of nosocomial infection in Madrid (Spain) 2006-2008 and later it was found that the functional inactivation of this two-partner system significantly reduces the adherence of bacteria to the A549 alveolar epithelial cell (Darvish Alipour Astaneh et al., 2014; Pérez et al., 2017; Sebastiaan J. van Hal et al., 2009). A channel-forming transporter/cytolysins activator of the TpsB family with a CDS length of 585 amino acids (WP_000248748.1) was also found in strain VRL-M19. Thus, the presence of these two partner systems in VRL-M19 indicates the possibility that it might also have remarkable adhesion capacity to the surfaces that facilitate the biofilm formation but further phenotypic exploration of this secretion system is required in VRL-M19.

3.3.4 Virulence factors in VRL-M19

The analysis for the virulence factors using the VFDB database revealed the presence of 54 virulence factors in VRL-M19, in addition to the one gene which was identified using the Victor database (PATRIC3.6.9) (**Supplementary Table 3**). The distribution of 55 virulence genes that were aligned on the ordered contigs of VRL-M19 is shown in **figure 4**. Among 55 virulence genes, 13 (*adeF*, *adeH*, *adeG*, *csuA*, *csuB*, *csuC*, *csuD*, *csuE*, *pgaA*, *pgaB*, *pgaC*, *pgaD*, *bap*) are known to be involved in biofilm formation and eight (*pgi*, *lpsB*, *lpxA*, *lpxB*, *lpxC*, *lpxD*, *lpxL*, *lpxM*) play role in immune evasion, nine virulence factors, namely, (genes ACICU_00071 (ATPase), ACICU_00072 (protein-tyrosine-phosphatase), ACICU_00073 (periplasmic protein), ACICU_00074 (UDP-N-acetyl-D-mannosaminuronate dehydrogenase), ACICU_00087 (sugar transferase), ACICU_00088 (UDP-glucose pyrophosphorylase), ACICU_00089 (UDP-glucose

6-dehydrogenase), **ACICU 00091** (UDP-glucose 4-epimerase), **ACICU 00092** (phosphomannomutase) and *pgi* are involved in immune invasion. A total of 16 genes (viz. *barB*, *basA*, *basB*, *basC*, *basF*, *basG*, *basH*, *basI*, *basJ*, *bauA*, *bauB*, *bauC*, *bauD*, *bauE*, *bauF*, *entE*) code for iron-chelating molecules called siderophores. Two genes, *bfmR*, and *bfmS*, are part of a two-component system thus and have a significant role in adhesion and biofilm formation. Two other genes, *abaI* and *abaR*, are involved in quorum sensing and *pbpG* in serum resistance. Among the virulence factors, *plc* and *plcD* encode for Phospholipase C and Phospholipase D enzymes respectively. The *ompA* gene encodes for surface proteins which are known to mediate pathogenesis via interaction with epithelial cells, resulting in cytotoxicity. Gene *ompA* is also known to be majorly involved in adherence and biofilm formation. The BRIG image illustrates the distribution of virulence factors on the genome of VRL-M19, revealing the presence of seven clusters of virulence genes. Notably, six genes (*picD*, *ompA*, *lpxB*, *lpxM*, *lpxC*, and *bap*) are dispersed across the genome. *OmpA* also plays crucial role in binding to human epithelial cells thus in adherence and biofilm formation (Matar et al., 2017) (Huang et al., 2016). Biofilm formation on abiotic surfaces involves the mediation of the pilus-related *csuA*, *csuB*, *csuC*, and *csuD* gene families, which are identified as annotated clusters in the VRL-M19 genome. (Pakharukova et al., 2013). The *A. baumannii* VRL-M19 was found to contain the *pgaABCD* gene cluster, known to encode the poly-Beta-(1-6) N-acetylglucosamine (PNAG) protein. PNAG serves as a crucial virulence factor, providing protection for bacteria against the host's innate immune response. The *A. baumannii* VRL-M19 isolate was found to contain the *pgaABCD* gene cluster, which is known to encode a protein called poly-Beta-(1-6) N-acetylglucosamine (PNAG). This protein serves as a crucial virulence factor, playing a significant role in protecting the bacteria from the innate immune response of the host. (Choi et al., 2009). The *adeF*, *adeG*

and *adeH* genes were detected as a cluster in the VRL-M19 genome, which were known to have correlation with the biofilm formation capabilities in the *A. baumannii* strains. The VRL-M19 genome harbors the largest cluster, comprising 16 genes, including *barB*, *basA*, *basB*, *basC*, *basF*, *basG*, *basH*, *basI*, *basJ*, *bauA*, *bauB*, *bauC*, *bauD*, *bauE*, *bauF*, and *entE*. This cluster is recognized for its involvement in the biosynthesis of acinetobactin, a siderophore iron acquisition system, and serves as a virulence factor crucial for pathogenesis in *A. baumannii*. (Conde-Pérez et al., 2021; Sheldon and Skaar, 2020). Furthermore, based on the prediction from the pathogen finder tool, strain VRL-M19 was identified as a human pathogen with 0.861 probability and contained 655 pathogenic protein families. These are 100 % similar to that of the other *A. baumannii* strains with the input proteome coverage of 18.29 % of VRL-M19. The predicted proteins mainly belong to type IV secretory protein IcmF, Sensory box protein, GasS-like sensor kinase protein, glycerol-3-Phosphate acetyltransferase, and a few hypothetical proteins that are known to be involved in the pathogenesis of *A. baumannii*. Based on the presence of pathogenic protein families and virulence gene clusters, strain VRL-M19 could be predicted as a potential human pathogen.

3.3.5 Drug resistance genes in VRL-M19

Total 69 drug resistance genes (18 genes from PRAP and 51 genes from CARD) were found in VRL-M19 genome (**Supplementary Table 4**). Most of the genes (30) belong to antibiotic efflux pump such as *emrE*, *adeL*, *adeF*, *adeG*, *adeH*, **WP_011859112.1** (code for ABC transporter ATP-binding protein/permease (obsolete from NCBI)), *acrB*, *envC*, *adeR*, *adeS*, **WP_011859328.1** (MFS transporter (obsolete from NCBI)), **WP_000128703.1** (*EmrA/EmrK* family), *emrB*, *lysR*, *craA*, *adeI*, *adeJ*, *adeK*, **WP_011860144.1** (*Bcr/CflA* family (obsolete from

NCBI), *adeN*, *macA*, ABO10989.2 (macrolide transport protein)), *abeM*, *tolC*, *adeB*, *adeA*, *mexK*, *macB*, *abeS*, *oqx*B while one gene i.e. *soxR*, belongs to both antibiotic efflux and target modification. There were 19 genes i.e. *gyrA*, *lpxC*, *lpxA*, *lpxD*, ABO12487.2, *fabG*, *parC*, *parE*, *rpsJ*, *ethA*, *folP*, *fabI*, *thyA*, *katG*, *rpoB*, *murA*, *rpsL*, *fusA*, *aadA* which belong to target modification. A total of six genes i.e. *carO*, *OprD*, **WP_000848135.1** (carbohydrate porin (obsolete from NCBI), *benP*, *quiX* and **WP_000910013.1** (*OprD* family outer membrane porin (obsolete from NCBI)) belong to permeability defects, and five belong to antibiotic inactivation, *adc-6*, *oxa-51*, *ant(3'')-II*, *aac(1)-I*, *add-A* while for eight genes i.e. *ftsK*, *estR*, *zapA*, *trm*, *recA*, *abrp*, *zipA* and *plsC*, mechanism of resistance is not clearly understood so far.

3.4. Pan-genome and mutational analysis of drug resistance genes

A pangenome analysis was conducted on 62 drug resistance genes of VRL-M19, comparing them with 788 global isolates and 350 Indian isolates. This analysis aimed to elucidate the distribution of core, accessory, and unique genes in relation to VRL-M19. During the analysis involving 788 clinical isolates, 51 drug resistance genes were identified as part of the core group, while 18 were categorized as accessory genes. Notably, no unique gene was observed in the VRL-M19 isolate. Likewise, in the analysis of 350 Indian isolates, a total of 17 drug-resistant genes were found in the core group, with 52 genes categorized as accessory. Similar to the VRL-M19 findings, no unique drug-resistant gene was identified among the Indian isolates. Given that VRL-M19 is an Indian isolate, it is noteworthy that the predominant resistance mechanism observed is antibiotic efflux, which is represented by a total of 30 genes. Among these, 18 were categorized as accessory genes, while 12 were identified as core genes.

Additionally, the pan-genome analysis performed on the 788 isolates, which are geographically

distributed worldwide, reveals the existence of 18 core genes and 12 accessory genes among the 30 drug resistance genes categorized under the antibiotic efflux resistance mechanism.

Among all accessory drug resistance genes, only *soxR* is characterized for conferring resistance via both antibiotic efflux and target modification resistance mechanisms (**Figure 5A**). It must be noted that all 18 accessory genes were significantly aligned with ≥ 740 clinical isolates of *A. baumannii* (**Figure 5B**).

The genome of VRL-M19 has 19 drug resistance target modification genes out of which 12 genes were found to contain mutations based on variation analysis carried out with respect to the reference genome of *A. baumannii* 17978-n^{cf} (**Table 2**). Among 12 genes, *parC* and *parE* are known for their resistance against fluoroquinolones and *folP* for sulphonamide, VRL-M19 also exhibits a phenotypic resistance to these classes of drugs. The antibiotic norfloxacin (fluoroquinolones class) is widely being prescribed and is known to act by inhibiting bacterial DNA replication by binding to the DNA gyrase (encoded by *gyrA* gene) and topoisomerase IV (encoded by *parC* and *parE* genes). Interestingly, in VRL-M19 several novel mutations are observed that might be associated with its phenotypic resistance. In *parC* three mutations, namely, Glu-457-->Lys, Gly-468-->Ser, and Thr-625-->Ala, and in *parE* one mutation at Val-551-->Ile, are observed.

A crystal structure of topoisomerase IV (**PDB ID: 2XKK**) of *A. baumannii* with moxifloxacin explains the structural basis of resistance which depict the H-bonding of Ser at 84 position and Glu at 88 positions of topoisomerase with moxifloxacin and thus mutations at these residues in topoisomerase IV confer quinolone resistance in gram-negative bacteria (Wohlkonig et al., 2010). In our analysis, a point mutation from acidic to the basic amino acid residue was also found in *parC* of VRL-M19 at 457 position (Glu-457-->Lys) which is not yet reported for

the quinolones resistance but may play a role in quinolone resistance. Further, in Pfam sequence analysis, *parC* mutations map to subunit A of gyrase/topoisomerase IV, indicating a potential role of these mutations in quinolone resistance. However, further studies are required to establish their role in drug resistance.

Furthermore, mutations were also observed in *folP* (encodes sulfonamide-targeted enzyme Dihydropteroate synthase (DHPS)). Analysis of *folP* of VRL-M19 for its resistance against Cotrimoxazole (SXT), a combination drug of sulfamethoxazole and trimethoprim, lead to the identification of 3 mutations (Arg-14-->Gly, Pro-130-->Ser, Met-280-->Ile) which were not reported earlier for *A. baumannii*. However, Met-280-->Ile mutation observed in VRL-M19 is reported in the literature for *Streptococcus* (Buwemba et al., 2013). Further, during the Pfam analysis of *folP* gene, Pro-130-->Ser mapped to the functional domain of the pterin binding enzyme while none of the other mutations mapped to any functional domains. It may be hypothesized that alterations in the pterin-binding enzyme due to this mutation could play a role in resistance against sulfonamides (Hevener et al., 2010; Patel et al., 2004). However, no reports of resistance to sulfonamides due to mutations in the pterin domain of the DHPS enzyme are found. It is important to note that there are several mechanisms of resistance to any antibiotic and only one such mechanism of target modification is explored in the current study.

In the pan-genome analysis of 788 isolates, a substantial portion of genes (51 out of 69) is designated as "Core," signifying their conservation across diverse bacterial strains. Conversely, Indian isolates exhibit a higher number of genes (52 out of 69) classified as "Accessory," suggesting potential variations in the gene pool or adaptations specific to the Indian context. Notably, a considerable number of genes classified as core in the 788 isolates are categorized as accessory in Indian isolates, indicating regional variations or specific adaptations. Comparative

analysis and visualization reveal a significant divergence in gene classification between the 788 global isolates and 350 Indian isolates. Such divergence could be crucial for understanding regional differences in drug resistance and tailoring antibiotic strategies accordingly.

3.5 Evaluation of *In-Vitro* pathogenesis of VRL-M19

3.5.1 Biofilm formation

The adherence capability of *A. baumannii* VRL-M19 to abiotic surfaces, specifically in terms of biofilm formation, was evaluated through the microtiter plate biofilm assay. The average of technical triplicates for both biological duplicates reveals that the absorbance at 595 nm is 2.738 for biological sample 1 and 2.680 for biological sample 2, falling within the range of 2 to 3. The findings indicate that *A. baumannii* VRL-M19 exhibited moderate biofilm-forming capabilities, suggesting a potential pathogenic nature for this strain (Supplementary Fig. 1).

3.5.2 Motility

In the swimming agar plate method, strain *A. baumannii* VRL-M19 was moved from the center of the agar plate to the periphery in 24 h of incubation. Whereas, this kind of motility was not observed for the nonmotile control strain *S. aureus* MTCC 1430. The visual observation of the relative movement of the culture from the center to the periphery of the plate determined that the strain VRL-M19 exhibited moderate twitching on-surface motility (Supplementary Fig. 2).

4. Conclusion:

Strain VRL-M19 isolated from a vegetable and fruit market, is an MDR *A. baumannii* that has

novel mutations in its drug resistance genes. Its genome analysis deciphers the presence of multiple clusters of virulence genes which were well reported in *A. baumannii* for its virulence and also predicted as a human pathogen. The Pan genome analysis indicated the substantial regional divergence in the distribution of drug resistance genes. The phylogeny and MLST analysis reveal its close relation with a clinical isolate of *A. baumannii*. The presence of Two Partner system (TPS) in the VRL-M19 genome is known for its biofilm formation and adherence properties. Evaluation of *In-Vitro* assays supported the pathogenic potential of the strain. Though the strain VRL-M19 was isolated from the environmental sample, the genomic features endorse its affiliation with clinical strains. The findings from the study would give a way to achieve further milestones in exploring the new drug targets and also to understand the genetic changes due to the transmission of virulent and drug-resistant determinants among the environmental strains that lead to the emergence of more MDR pathogenic strains.

Nucleotide sequence accession no. and culture availability: The genome sequence and 16S rRNA gene sequences of *A. baumannii* VRL-M19 have been deposited to GenBank under accession no. **JAHLEI000000000.1** and **OQ692086** respectively. *A. baumannii* strain VRL-M19 is available at the Microbial Type Culture Collection (MTCC) and Gene Bank, India with accession number **MTCC 13285**.

Funding: Not funded by any grant. This study was supported by institute's basic facilities from CSIR-Institute of Microbial Technology, Sector-39A, Chandigarh. The institutional manuscript communication number of the manuscript is 060/2021.

Ethical approval: Not required

Competing interests: None declared

Acknowledgments: Authors acknowledge MTCC & Director, CSIR-IMTECH (Host Institute) for providing basic requirements & infrastructure for this study.

Data availability statement: The details of the sequence data available online is given in the manuscript; further enquiries may be directed to the corresponding author.

References:

- Alarcon, I., Evans, D.J., Fleiszig, S.M.J., 2002. The Role of Twitching Motility in *Pseudomonas aeruginosa* Exit from and Translocation of Corneal Epithelial Cells. *Invest Ophthalmol Vis Sci* 50, 2237. <https://doi.org/10.1167/IOVS.08-2785>
- Alikhan, N.F., Petty, N.K., Ben Zakour, N.L., Beatson, S.A., 2011. BLAST Ring Image Generator (BRIG): Simple prokaryote genome comparisons. *BMC Genomics* 12,402. <https://doi.org/10.1186/1471-2164-12-402>
- Almaghrabi, M.K., Joseph, M.R.P., Assiry, M.M., Hamid, M.E., 2018. Multidrug-Resistant *Acinetobacter baumannii*: An Emerging Health Threat in Aseer Region, Kingdom of Saudi Arabia. *Can J Infect Dis Med Microbiol* 2018. <https://doi.org/10.1155/2018/9182747>
- Antonopoulos, D.A., Assaf, R., Aziz, R.K., Brettin, T., Bun, C., Conrad, N., Davis, J.J., Dietrich, E.M., Disz, T., Gerdes, S., Kenyon, R.W., Machi, D., Mao, C., Murphy-Olson, D.E., Nordberg, E.K., Olsen, G.J., Olson, R., Overbeek, R., Parrello, B., Pusch, G.D., Santerre, J., Shukla, M., Stevens, R.L., Vanoeffelen, M., Vonstein, V., Warren, A.S., Wattam, A.R., Xia, F., Yoo, H., 2019. PATRIC as a unique resource for studying antimicrobial resistance. *Brief Bioinform* 20, 1094–1102. <https://doi.org/10.1093/BIB/BBX083>
- Badave, G.K., Dhananjay, K., 2015. Biofilm Producing Multidrug Resistant *Acinetobacter*

- baumannii*: An Emerging Challenge. J Clin Diagn Res 9, DC08.
<https://doi.org/10.7860/JCDR/2015/11014.5398>
- Banerjee, T., Mishra, A., Das, A., Sharma, S., Barman, H., Yadav, G., 2018. High Prevalence and Endemicity of Multidrug Resistant *Acinetobacter* spp. in Intensive Care Unit of a Tertiary Care Hospital, Varanasi, India . J Pathog 2018, 1–8.
<https://doi.org/10.1155/2018/9129083>
- Bortolaia, V., Kaas, R.S., Ruppe, E., Roberts, M.C., Schwarz, S., Cattoir, V., Philippon, A., Allesoe, R.L., Rebelo, A.R., Florensa, A.F., Fagelhauer, L., Chakraborty, T., Neumann, B., Werner, G., Bender, J.K., Stingl, K., Nguyen, M., Coppens, J., Xavier, B.B., Malhotra-Kumar, S., Westh, H., Pinholt, M., Anjum, M.F., Duggett, M.A., Kempf, I., Nykäsenoja, S., Olkkola, S., Wieczorek, K., Amaro, A., Clemente, L., Messong, J., Losch, S., Ragimbeau, C., Lund, O., Aarestrup, F.M., 2020. ResFinder 4.0 for predictions of phenotypes from genotypes. Journal of Antimicrobial Chemotherapy 75, 3491–3500.
<https://doi.org/10.1093/JAC/DKAA345>
- Brettin, T., Davis, J.J., Disz, T., Edwards, R.A., Cerdes, S., Olsen, G.J., Olson, R., Overbeek, R., Parrello, B., Pusch, G.D., Shukla, M., Thomason, J.A., Stevens, R., Vonstein, V., Wattam, A.R., Xia, F., 2015. RASTtk: A modular and extensible implementation of the RAST algorithm for building custom annotation pipelines and annotating batches of genomes. Scientific Reports 2015 5:1 5, 1–6. <https://doi.org/10.1038/srep08365>
- Buwembo, W., Aery, S., Rwenyonyi, C.M., Swedberg, G., Kironde, F., 2013. Point mutations in the folP gene partly explain sulfonamide resistance of streptococcus mutans. Int J Microbiol 2013, 367021. <https://doi.org/10.1155/2013/367021>
- Chaudhari, N.M., Gupta, V.K., Dutta, C., 2016. BPGA- an ultra-fast pan-genome analysis pipeline. Scientific Reports 2016 6:1 6, 1–10. <https://doi.org/10.1038/srep24373>
- Chen, L., Zheng, D., Liu, B., Yang, J., Jin, Q., 2016. VFDB 2016: hierarchical and refined dataset for big data analysis--10 years on. Nucleic Acids Res 44, D694–D697.
<https://doi.org/10.1093/NAR/GKV1239>
- Choi, A.H.K., Slamti, L., Avci, F.Y., Pier, G.B., Maira-Litrán, T., 2009. The pgaABCD locus of *Acinetobacter baumannii* encodes the production of poly-β-1-6-N-acetylglucosamine, which is critical for biofilm formation. J Bacteriol 191, 5953–5963.
<https://doi.org/10.1128/JB.00647-09/ASSET/41A250E7-BDBF-4297-8A0D->

C96BA530CE9A/ASSETS/GRAPHIC/ZJB0190990360007.JPEG

- Conde-Pérez, K., Vázquez-Ucha, J.C., Álvarez-Fraga, L., Ageitos, L., Rumbo-Feal, S., Martínez-Gutián, M., Trigo-Tasende, N., Rodríguez, J., Bou, G., Jiménez, C., Beceiro, A., Poza, M., 2021. In-Depth Analysis of the Role of the Acinetobactin Cluster in the Virulence of *Acinetobacter baumannii*. *Front Microbiol* 12, 3010.
<https://doi.org/10.3389/FMICB.2021.752070/BIBTEX>
- Cosentino, S., Voldby Larsen, M., Møller Aarestrup, F., Lund, O., 2013. PathogenFinder - Distinguishing Friend from Foe Using Bacterial Whole Genome Sequence Data. *PLoS One* 8, e77302. <https://doi.org/10.1371/JOURNAL.PONE.0077302>
- da Silva Nali, L.H., Fujita, D.M., Salvador, F.S., da Silva Fink, M.C.D., de Andrade, H.F., Pannuti, C.S., de Albuquerque Luna, E.J., 2016. Potential measles transmission risk in mass gatherings: Are we safe for the Olympic games-Rio 2016? *J Travel Med* 23, 1–3.
<https://doi.org/10.1093/jtm/taw026>
- Darvish Alipour Astaneh, S., Rasooli, I., Mousavi Gargari, S.L., 2014. The role of filamentous hemagglutinin adhesin in adherence and biofilm formation in *Acinetobacter baumannii* ATCC19606T. *Microb Pathog Complex*, 42–49.
<https://doi.org/10.1016/J.MICPATH.2014.07.007>
- Eijkkelkamp, B.A., Stroehner, U.H., Høsten, K.A., Papadimitriou, M.S., Paulsen, I.T., Brown, M.H., 2011. Adherence and motility characteristics of clinical *Acinetobacter baumannii* isolates. *FEMS microbiology letters*, 323(1), 44–51. <https://doi.org/10.1111/j.1574-6968.2011.02362.x>
- Ellington, M.J., Ekelund, C., Aarestrup, F.M., Canton, R., Doumith, M., Giske, C., Grundman, H., Hasman, H., Heuvelink, M.T.G., Hopkins, K.L., Iredell, J., Kahlmeter, G., Köser, C.U., MacGowan, A., Mevius, D., Mulvey, M., Naas, T., Peto, T., Rolain, J.M., Samuelsen, Woodford, N., 2017. The role of whole genome sequencing in antimicrobial susceptibility testing of bacteria: report from the EUCAST Subcommittee. *Clinical Microbiology and Infection* 23, 2–22. <https://doi.org/10.1016/J.CMI.2016.11.012>
- Feldgarden, M., Brover, V., Haft, D.H., Prasad, A.B., Slotta, D.J., Tolstoy, I., Tyson, G.H., Zhao, S., Hsu, C.H., McDermott, P.F., Tadesse, D.A., Morales, C., Simmons, M., Tillman, G., Wasilenko, J., Folster, J.P., Klimke, W., 2019. Validating the AMRFINDER tool and resistance gene database by using antimicrobial resistance genotype-phenotype correlations

- in a collection of isolates. Antimicrob Agents Chemother 63.
<https://doi.org/10.1128/AAC.00483-19>
- Finn, R.D., Bateman, A., Clements, J., Coggill, P., Eberhardt, R.Y., Eddy, S.R., Heger, A., Hetherington, K., Holm, L., Mistry, J., Sonnhammer, E.L.L., Tate, J., Punta, M., 2014. Pfam: The protein families database. Nucleic Acids Res 42, 222–230.
<https://doi.org/10.1093/nar/gkt1223>
- Gautret, P., Steffen, R., 2016. Communicable diseases as health risks at mass gatherings other than Hajj: What is the evidence? International Journal of Infectious Diseases 47, 46–52.
<https://doi.org/10.1016/j.ijid.2016.03.007>
- Gedefie, A., Demsis, W., Ashagrie, M., Kassa, Y., Tesfaye, M., Filahun, M., Bisetegn, H., Sahle, Z., 2021. *Acinetobacter baumannii* biofilm formation and its role in disease pathogenesis: A review. Infect Drug Resist 14, 3711–3719. <https://doi.org/10.2147/IDR.S332051>
- Han, X., Kennan, R.M., Davies, J.K., Reddacliff, L.A., Dhangyel, O.P., Whittington, R.J., Turnbull, L., Whitchurch, C.B., Rood, J.I., 2008. Twitching motility is essential for virulence in *Dichelobacter nodosus*. J Bacteriol 190, 3323–3335.
<https://doi.org/10.1128/JB.01807-07>
- Harding, C.M., Tracy, E.N., Carruthers, M.D., Rather, P.N., Actis, L.A., Munson, R.S., Taylor, R., 2013. *Acinetobacter baumannii* Strain M2 Produces Type IV Pili Which Play a Role in Natural Transformation and Twitching Motility but Not Surface-Associated Motility 4 (4), e00360-13. <https://doi.org/10.1128/mBio.00360-13>
- He, Y., Zhou, X., Chen, Z., Deng, X., Gehring, A., Ou, H., Zhang, L., Shi, X., 2020. PRAP: Pan Resistome analysis pipeline. BMC Bioinformatics 21, 1–8. <https://doi.org/10.1186/S12859-019-3335-Y/FIGURES/5>
- Hevener, K.E., Yun, M.K., Qi, J., Kerr, I.D., Babaoglu, K., Hurdle, J.G., Balakrishna, K., White, S.W., Lee, R.E., 2010. Structural Studies of Pterin-Based Inhibitors of Dihydropteroate Synthase. J Med Chem 53, 166. <https://doi.org/10.1021/JM900861D>
- Huang, W., Yao, Yufeng, Wang, S., Xia, Y., Yang, X., Long, Q., Sun, W., Liu, C., Li, Y., Chu, X., Bai, H., Yao, Yueting, Ma, Y., 2016. Immunization with a 22-kDa outer membrane protein elicits protective immunity to multidrug-resistant *Acinetobacter baumannii*. Sci Rep 6. <https://doi.org/10.1038/srep20724>
- Jia, B., Raphenya, A.R., Alcock, B., Wagtelechner, N., Guo, P., Tsang, K.K., Lago, B.A., Dave,

- B.M., Pereira, S., Sharma, A.N., Doshi, S., Courtot, M., Lo, R., Williams, L.E., Frye, J.G., Elsayegh, T., Sardar, D., Westman, E.L., Pawlowski, A.C., Johnson, T.A., Brinkman, F.S.L., Wright, G.D., McArthur, A.G., 2017. CARD 2017: expansion and model-centric curation of the comprehensive antibiotic resistance database. *Nucleic Acids Res* 45, D566–D573. <https://doi.org/10.1093/NAR/GKW1004>
- Larsen, M. V., Cosentino, S., Rasmussen, S., Friis, C., Hasman, H., Marvig, R.L., Jelsbak, L., Sicheritz-Pontén, T., Ussery, D.W., Aarestrup, F.M., Lund, O., 2012. Multilocus sequence typing of total-genome-sequenced bacteria. *J Clin Microbiol* 50, 1355–1361. <https://doi.org/10.1128/JCM.06094-11>
- Leangapichart, T., Gautret, P., Griffiths, K., Belhouchat, K., Memish, Z., Raoult, D., Rolain, J.M., 2016. Acquisition of a high diversity of bacteria during the Hajj pilgrimage, including *Acinetobacter baumannii* with blaOXA-72 and *Escherichia coli* with blaNDM-5 carbapenemase genes. *Antimicrob Agents Chemother* 60, 5942–5948. <https://doi.org/10.1128/AAC.00669-16>
- Matar, G.M., Sun Yoon, S., University, Y., Korea Ravi Jhaveri, S., Lee, C.-R., Hun Lee, J., Park, M., Seung Park, K., Kwon Bae, I., Bae Kim, Y., Cha, C.-J., Chul Jeong, B., Hee Lee, S., 2017. Biology of *Acinetobacter baumannii*: Pathogenesis, Antibiotic Resistance Mechanisms, and Prospective Treatment Options. *Frontiers in Cellular and Infection Microbiology* | www.frontiersin.org 7, 55. <https://doi.org/10.3389/fcimb.2017.00055>
- Memish, Z.A., Steffen, R., White, P., Dar, O., Azhar, E.I., Sharma, A., Zumla, A., 2019. Mass gatherings medicine: public health issues arising from mass gathering religious and sporting events. *Lancet* 393, 2075. [https://doi.org/10.1016/S0140-6736\(19\)30501-X](https://doi.org/10.1016/S0140-6736(19)30501-X)
- Mikheenko, A., Prjibelski, A., Saveliev, V., Antipov, D., Gurevich, A., 2018. Versatile genome assembly evaluation with QUAST-LG, in: *Bioinformatics*. Oxford University Press, pp. i142–i150. <https://doi.org/10.1093/bioinformatics/bty266>
- Page, A.J., Taylor, B., Delaney, A.J., Soares, J., Seemann, T., Keane, J.A., Harris, S.R., 2016. SNP-sites: rapid efficient extraction of SNPs from multi-FASTA alignments. *Microb Genom* 2, e000056. <https://doi.org/10.1099/MGEN.0.000056/CITE/REFWORKS>
- Pakharukova, N., Tuittila, M., Paavilainen, S., Malmi, H., Parilova, O., Teneberg, S., Knight, S.D., Zavialov, A. V., 2018. Structural basis for *Acinetobacter baumannii* biofilm formation. *Proc Natl Acad Sci U S A* 115, 5558–5563.

- https://doi.org/10.1073/PNAS.1800961115/SUPPL_FILE/PNAS.1800961115.SAPP.PDF
- Patel, O.G., Mberu, E.K., Nzila, A.M., Macreadie, I.G., 2004. Sulfa drugs strike more than once. *Trends Parasitol* 20, 1–3. <https://doi.org/10.1016/J.PT.2003.10.009>
- Pérez, A., Merino, M., Rumbo-Feal, S., Álvarez-Fraga, L., Vallejo, J.A., Beceiro, A., Ohneck, E.J., Mateos, J., Fernández-Puente, P., Actis, L.A., Poza, M., Bou, G., 2017. The FhaB/FhaC two-partner secretion system is involved in adhesion of *Acinetobacter baumannii* AbH12o-A2 strain. *Virulence* 8, 959–974. <https://doi.org/10.1080/21505594.2016.1262313>
- Rissman, A.I., Mau, B., Biehl, B.S., Darling, A.E., Glasner, J.D., Perna, N.T., 2009. Reordering contigs of draft genomes using the Mauve Aligner. *Bioinformatics* 25, 2071. <https://doi.org/10.1093/BIOINFORMATICS/BTP356>
- Santajit, S., Indrawattana, N., 2016. Mechanisms of Antimicrobial Resistance in ESKAPE Pathogens. *Biomed Res Int* 2016. <https://doi.org/10.1155/2016/2475067>
- Sayers, E.W., Bolton, E.E., Brister, J.R., Canese, K., Chan, J., Comeau, D.C., Connor, R., Funk, K., Kelly, C., Kim, S., Madej, T., Marchler-Bauer, A., Lanczycki, C., Lathrop, S., Lu, Z., Thibaud-Nissen, F., Murphy, T., Phan, T., Skripchenko, Y., Tse, T., Wang, J., Williams, R., Trawick, B.W., Pruitt, K.D., Sherry, S.T., 2022. Database resources of the national center for biotechnology information. *Nucleic Acids Res* 50, D20–D26. <https://doi.org/10.1093/nar/glab112>
- Sharma, T., Kumar, R., Bhardwaj, A., n.d. Ab-AMR: A Comprehensive Repository of *Acinetobacter baumannii* to Understand the Molecular Landscape of Antimicrobial Resistance. <https://doi.org/10.1101/2022.07.17.500328>
- Sharma, R.K., Mamoria, V.P., Author, C., 2018. A Prospective Study on Prevalence and Antibiotic Susceptibility Pattern of *Acinetobacter baumannii* in Clinical Samples obtained from Patients admitted in Various Wards and Intensive Care Units. *Journal of Mahatma Gandhi University of Medical Sciences and Technology* 2, 122–127. <https://doi.org/10.5005/JP-JOURNALS-10057-005>
- Sheldon, J.R., Skaar, E.P., 2020. *Acinetobacter baumannii* can use multiple siderophores for iron acquisition, but only acinetobactin is required for virulence. *PLoS Pathog* 16, e1008995. <https://doi.org/10.1371/JOURNAL.PPAT.1008995>
- Singh, H., Thangaraj, P., Chakrabarti, A., 2013b. *Acinetobacter baumannii*: A brief account of

- mechanisms of multidrug resistance and current and future therapeutic management. *Journal of Clinical and Diagnostic Research* 7, 2602–2605.
<https://doi.org/10.7860/JCDR/2013/6337.3626>
- Taneja, N., Sharma, M., 2019. Antimicrobial resistance in the environment: The Indian scenario. *Indian Journal of Medical Research* 149,119–128.
https://doi.org/10.4103/ijmr.IJMR_331_18
- Tenover, F.C., 2019. Antimicrobial Susceptibility Testing. *Encyclopedia of Microbiology* 166–175. <https://doi.org/10.1016/B978-0-12-801238-3.02486-7>
- Thompson, J.D., Higgins, D.G., Gibson, T.J., 1994. CLUSTAL W: Improving the sensitivity of progressive multiple sequence alignment through sequence weighting, position-specific gap penalties and weight matrix choice, *Nucleic Acids Research* 22(22), 4673–4680.
<https://doi.org/10.1093/nar/22.22.4673>
- van Hal, Sebastiaan J., Foo, H., Blyth, C.C., McPhie, K., Armstrong, P., Sintchenko, V., Dwyer, D.E., 2009. Influenza outbreak during Sydney World Youth Day 2008: The utility of laboratory testing and case definitions on mass gathering outbreak containment. *PLoS One* 4. <https://doi.org/10.1371/journal.pone.0006620>
- Vijayakumar, S., Mathur, P., Kapil, A., Das, B.K., Ray, P., Gautam, V., Sistla, S., Parija, S.C., Walia, K., Ohri, V.C., Anandan, S., Subramani, K., Ramya, I., Veeraraghavan, B., 2019. Molecular characterization & epidemiology of carbapenem-resistant *Acinetobacter baumannii* collected across India. *Indian J Med Res* 149, 240.
https://doi.org/10.4103/IJMR.IJMR_2085_1
- Vrancianu, C.O., Ghicoghici, L., Czobor, I.B., Chifiriuc, M.C., 2020. Antibiotic resistance profiles, molecular mechanisms and innovative treatment strategies of *Acinetobacter baumannii*. *Microorganisms* 8, 1–40. <https://doi.org/10.3390/microorganisms8060935>
- Weinstein, M.P., Clinical and Laboratory Standards Institute, n.d. Performance standards for antimicrobial susceptibility testing.
- Weisburg, W.G., Barns, S.M., Pelletier, D.A., Lane, D.J., 1991. 16S ribosomal DNA amplification for phylogenetic study. *J Bacteriol* 173, 697.
<https://doi.org/10.1128/JB.173.2.697-703.1991>
- Wharton, M., Spiegel, R.A., Horan, J.M., Tauxe, R. v, Wells, J.G., Barg, N., Herndon, J., Meriwether, R.A., MacCormack, J.N., Levine, R.H., 1990. A large outbreak of antibiotic-

- resistant shigellosis at a mass gathering. *Journal of Infectious Diseases* 162, 1324–1328.
<https://doi.org/10.1093/infdis/162.6.1324>
- Wohlkonig, A., Chan, P.F., Fosberry, A.P., Homes, P., Huang, J., Kranz, M., Leydon, V.R., Miles, T.J., Pearson, N.D., Perera, R.L., Shillings, A.J., Gwynn, M.N., Bax, B.D., 2010. Structural basis of quinolone inhibition of type IIA topoisomerases and target-mediated resistance. *Nat Struct Mol Biol* 17, 1152–1153. <https://doi.org/10.1038/nsmb.1892>
- Xie, Z., Tang, H., 2017. ISEScan: automated identification of insertion sequence elements in prokaryotic genomes. *Bioinformatics* 33, 3340–3347.
<https://doi.org/10.1093/bioinformatics/btx433>
- Yadav, S.K., Bhujel, R., Hamal, P., Mishra, S.K., Sharma, S., Shrestha, J.B., 2020. Burden of Multidrug-Resistant *Acinetobacter baumannii* Infection in Hospitalized Patients in a Tertiary Care Hospital of Nepal. *Infect Drug Resist* 13, 725–732.
<https://doi.org/10.2147/IDR.S239514>
- Yoon, S.H., Ha, S.M., Kwon, S., Lim, J., Kim, Y., Seo, H., Chun, J., 2017. Introducing EzBioCloud: A taxonomically united database of 16S rRNA gene sequences and whole-genome assemblies. *Int J Syst Evol Microbiol* 67, 1613–1617.
<https://doi.org/10.1099/ijsem.0.001755>
- Zeighami, H., Valadkhani, F., Shapouri, E., Samadi, E., Haghi, F., 2019. Virulence characteristics of multidrug resistant biofilm forming *Acinetobacter baumannii* isolated from intensive care unit patients. *BMC Infect Dis* 19(1), 629.
<https://doi.org/10.1186/s12979-019-4272-0>
- Zhou, H., Gebhardt, M.J., Czyz, D.M., Yao, Y., Shuman, H.A., 2021. The *gigA/gigB* Genes Regulate the Growth, Stress Response, and Virulence of *Acinetobacter baumannii* ATCC 17978 Strain. *Front Microbiol* 12. <https://doi.org/10.3389/fmicb.2021.723949>

Journal Pre-proof

Table 1. The phenotypic resistance and susceptible profile exhibited by *A. baumannii* VRL-M19.

Antibiotic	Class	Resistance profile
Oxacillin (1mcg)	Beta -Lactam	+
Cephalothin (30mcg)		+
Co-trimoxazole (25mcg)	Sulphonamide	+
Erythromycin (15mcg)	Macrolide	+
Norfloxacin (10mcg)	Fluoroquinolones	+
Nitrofurantoin (300mcg)	Nitrofurantoin	+
Aztreonam (30mcg)	Monobactam	+
Cefadroxil (30mcg)	Cephalosporin	+
Ampicillin (10mcg)	Beta-Lactam	-
Cefotaxime (30mcg)		-
Amoxicillin (10mcg)		-
Cefuroxime (30mcg)		-
Ceftazidime (30mcg)		-
Piperacillin (100mcg)		-
Ticarcillin (75mcg)		-
Carbenicillin (100mcg)		-
Gentamicin (10mcg)	Aminoglycoside	-
Neomycin (30mcg)		-
Streptomycin (10mcg)		-

Amikacin (30mcg)		-
Tobramycin (10mcg)		-
Netillin (10mcg)		-
Tetracycline (25mcg)	Tetracycline	-
Doxycycline (30mcg)		-
Sulphatriad (300mcg)	Sulphonamide	-
Ciprofloxacin (5mcg)	fluoroquinolones	-
Ofloxacin (5mcg)		-
Gatifloxacin (5mcg)		-
Imipenem (10mcg)	Carbapenem	-
Meropenem (10mcg)		-
Teicoplanin (30mcg)	Glycopeptide	-
Cefoperazone/Sulbactam (75/10mcg)	Antibiotics-Adjuvant	-
Piperacillin/Tazobactam (100/10mcg)		-
Ampicillin/Sulbactam (10/10mcg)		-
Ticarcillin/Calvulanic (75/10mcg)		-

+, Resistance; -, Susceptible, Concentration of antibiotics used are given in the parenthesis.

Table 2. List of mutations in drug resistance genes in the VRL-M19 genome in the context of the Pfam domain and their functions

Gene	Antibiotics (class)	Mutation positions	Pfam (sequence search)	Accession_id (Pfam)
<i>*lpxC</i>	Colistin (Polymyxin B)	Cys-120-- >Arg	UDP-3-O-acylN-acetylglucosamine deacetylase	PF03331
<i>lpxA</i>	Colistin (Polymyxin B)	Tyr-131-- >His	Bacterial heptapeptide transferase	PF00132
<i>lpxD</i>	Colistin (Polymyxin B)	Val-3-->Ala	Not mapped	
<i>parC</i>	Nybomycin (Fluoroquinolone)	Glu-457-- >Lys, Gly-468-- >Ser, The-625-- >Ala	DNA gyrase/topoisomerase IV, subunit A	PF00521
<i>folP</i>	(Sulfonamide antibiotic, sulfone antibiotic)	Arg-14-- >Gly, Pro-130-- >Ser, Met- 280-- >Ile	Not Mapped Not Mapped Pterin binding enzyme	PF00809

<i>parE</i>	(Fluoroquinolone)	Val-55--> Ile	Not mapped	
<i>fabG</i>	Triclosan	Ser-272-- >Gly, Asp-344-- >Gly	Enoyl-(Acyl-carrier protein) reductase	PF13561
<i>ethA</i>	Isoniazid, Ethionamide	Thr-488-- >Ser	Not mapped	
<i>fabI</i>	Isoniazid, Triclosan	Glu-65-- >Gln, Glu-129-- >Asp	Enoyl- (Acyl-carrier protein) reductase	PF13561
<i>thyA</i>	Para-aminosalicylic acid	Asp-211-- >Gly, Asp-251-- >Glu	Thymidylate synthase	PF00303
<i>katG</i>	Isoniazid	Pro-270-- >Ser, Ala-457-- >Thr, His-484-- >Arg, Gln-505--	Not Mapped Not Mapped Not Mapped Peroxidase	PF00141

		>Lys		
<i>rpoB</i>	Peptide antibiotic, Rifamycin antibiotic	Ile-604-- >Val	RNA polymerase beta subunit external 1 domain	PF10385

* Except for Cys-120-->Arg mutation of *lpxC* gene, the rest of the mutations are not reported for

A. baumannii in the literature

Figure 1. Genome comparisons of *A. baumannii* VRL-M19 and *A. baumannii* ATCC 17978-mff.

Genome comparisons were done using contig assemblies showing homologous regions with closely related reference genome of *A. baumannii* ATCC 17978-mff (A). The 19 LCB plot was generated with the contig reordering tool MCM within Mauve by using default parameters. The strain name is listed to on the left of side of each LCB plot. The contigs contained within the VRL-M19 (B) draft genome are reordered to approximate the LCB plot of the complete genome of *A. baumannii* ATCC 17978-mff.

Figure 2. Whole genome phylogenetic tree depicting the evolutionary relationship of VRL-M19 to other *A. baumannii* strains of clinical and environmental origin.

Number of nodes represent bootstrap values (1000 reiterations). Bootstrap values $\geq 50\%$ were indicated at the branch points. Bar, 0.05 substitution for 100 nucleotide positions. The sequence accession numbers are given in parentheses.

Figure 3. Subsystem analysis of *A. baumannii* VRL-M19 genome.

Subsystem distribution determined by RAST SEED analysis. The pie chart depicts the subsystems corresponds to cellular process and number of respective protein-coding genes (in parentheses) of strain VRL-M19.

Figure 4. Distribution of the virulence factors is shown using BRIG.

The innermost concentric circle (black) is the concatenated ordered contigs of VRL-M19. The subsequent circles are GC content (black) and GC skew (+ (purple) and - (green)), the red circle representing the CDS and the outermost black circle with labels representing the virulence

factors of VRL-M19. Gene1 is corresponding to (WP_002055229.1). Likewise, Gene2 (WP_001165095.1), Gene3 (WP_000872593.1), Gene4 (WP_000209960.1), Gene5 (WP_001062908.1), Gene6 (WP_000686134.1), Gene7 (WP_000591436.1), Gene8 (WP_001165990.1), Gene9 (WP_001075965.1).

Figure 5. Pan-genome analysis of 69 drug resistance genes

(A) The distribution of 69 DR determinants. Out of 69, 51 genes belong to the core and 18 genes are accessory. There is one gene viz. *soxR* which shows both antibiotic efflux and target modification mechanism of resistance **(B)** The 18 accessory drug resistance genes of VRL-M19 and its distribution among all clinical isolates of *A. baumannii*.

Supplementary Table 1. The susceptibility profile exhibited by *A. baumannii* VRL-M19 against various antibiotics their combinations

Supplementary Table 2. Sequence Typing of VRL-M19 strain based on Pasture scheme

Supplementary Table 3. The Sequence Typing of VRL-M19 strain based on the Oxford scheme

Supplementary Table 4. Virulence genes of *Acinetobacter baumannii* VRL-M19 genome

Supplementary Table 5. Drug-resistance genes of *Acinetobacter baumannii* VRL-M19 genome

Supplementary Table 6. Details and distribution drug resistance genes of global and Indian clinical strains of *A. baumannii* used for pan genome analysis.

Worksheet 1 of Supplementary Table 6. Comprehensive list of 350 high-quality genome isolates from India, specifically employed for pan-genome analysis.

Worksheet 2 of Supplementary Table 6. Lists of the 788 clinical isolates used for pan-

genome analysis.

Worksheet 3 of Supplementary Table 6. Geographical location of 728 global isolates (out of 788) available in the PATRIC.

Worksheet 4 of Supplementary Table 6. Full name of the 69 drug-resistance (DR) genes identified in the VRL-M19 isolate.

Worksheet 5 of Supplementary Table 6. Comprehensive list and alignment of 69 drug-resistance genes with global and Indian isolates.

Supplementary Figure 1. Biofilm formation by *A. baumannii* VRL-M19 in the microtiter plate assay.

Supplementary Figure 2. *A. baumannii* VRL-M19 showing the twitching on-surface motility on the LB agar plate.

Data statement

Nucleotide sequence accession no. and culture availability: The genome sequence and 16S rRNA gene sequences of *A. baumannii* VRL-M19 have been deposited to GenBank under accession no. **JAHLEI000000000.1** and **OQ692086** respectively. Strain *Acinetobacter baumannii* VRL-M19 is available at the Microbial Type Culture Collection (MTCC) and Gene Bank, India with accession number **MTCC 13285**.

Any further enquiries related to data may be directed to the corresponding author at venkat@imtech.res.in

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Highlights of the study

Title: Comparative genomics and characterization of a multidrug-resistant *Acinetobacter baumannii* VRL-M19 isolated from a crowded setting in India

The study reports the isolation of the multidrug-resistant *Acinetobacter baumannii* strain VRL-M19 from the air of a crowded vegetable market, showcasing significant drug-resistant genomic features aligned with its phenotypic characteristics. The research involves comprehensive phenotypic and genotypic characterization, along with a comparative genomic analysis of 788 *A. baumannii* strains, along with the one isolated from the current air sample. Novel genetic mutations related to drug-resistant determinants and identified virulence genes associated with airborne *A. baumannii* suggest its potential as a human pathogen. Molecular phylogeny supports the clinical origin prediction of strain VRL-M19, despite its isolation from the environmental (air) setting. The genetic and *in-vitro* features of the strain justify its biofilm formation capacity. Pangenome analysis of 788 global isolates and 300 Indian isolates of *A. baumannii* exhibited a substantial divergence in gene classification, providing key insights into regional disparities in drug resistance. This study holds importance in addressing antimicrobial resistance (AMR), offering insights into new drug targets, and understanding genetic changes contributing to the transmission of virulent and drug-resistance determinants among environmental strains, leading to the emergence of more multidrug-resistant (MDR) pathogenic strains.

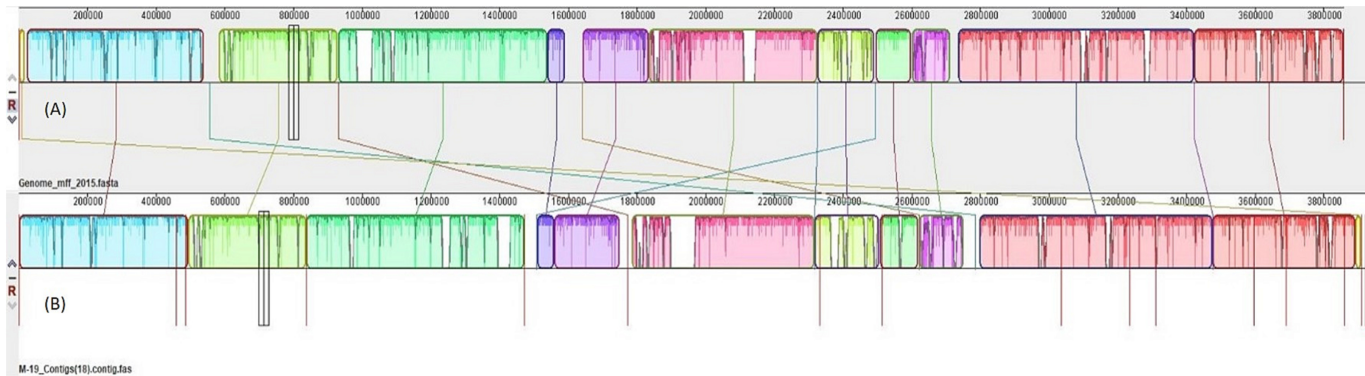


Figure 1

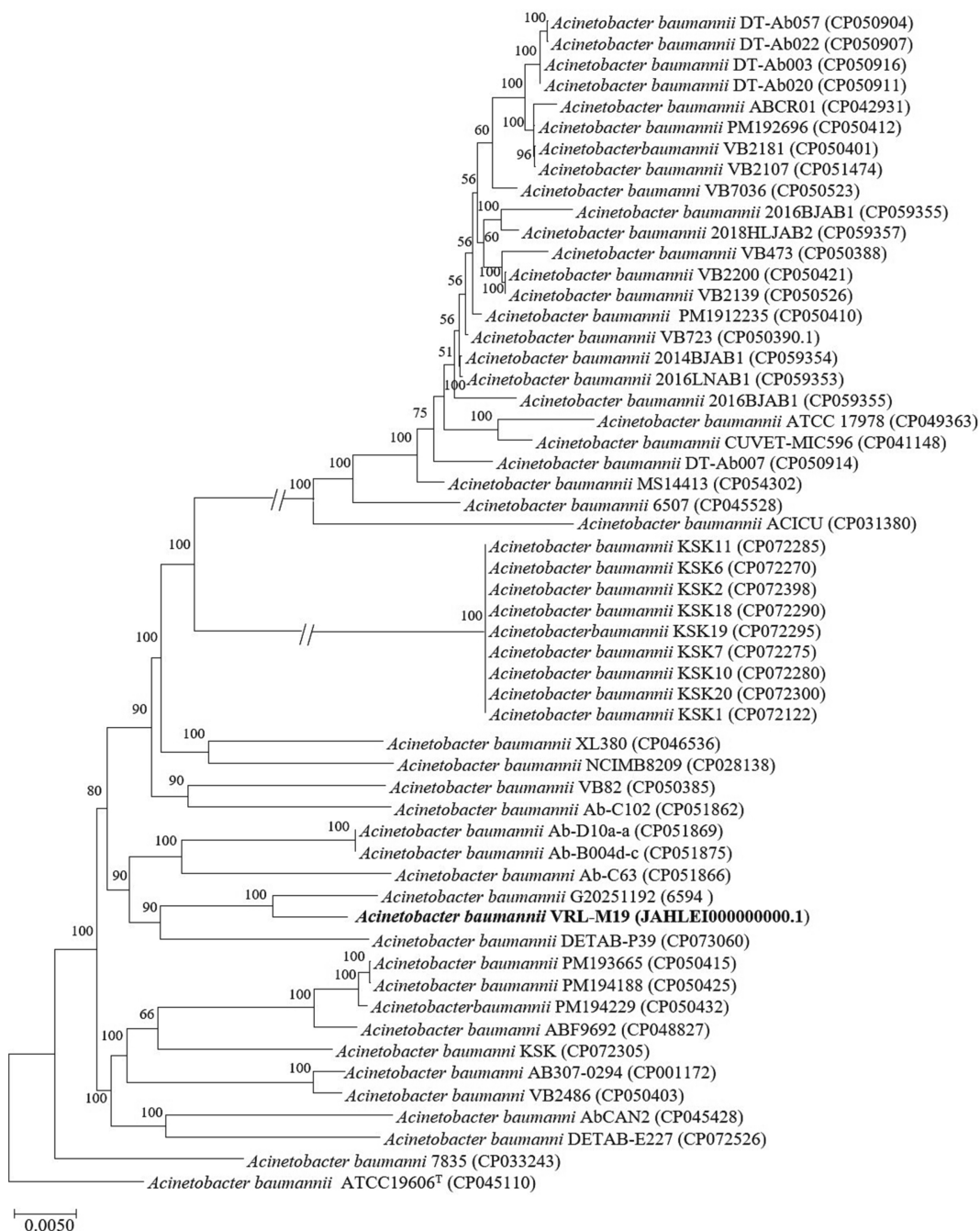
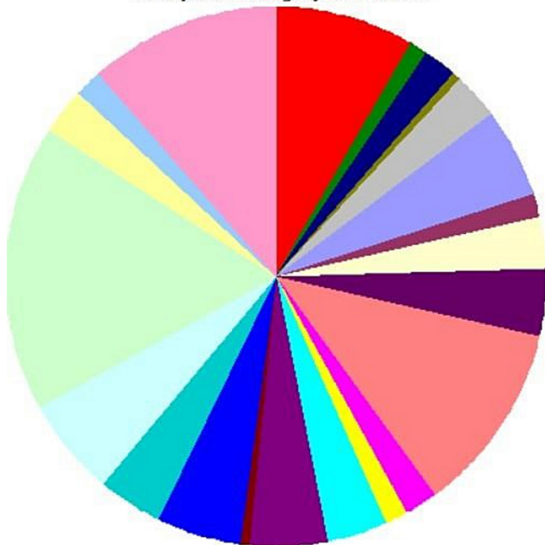


Figure 2

Subsystem Category Distribution



Subsystem Feature Counts

- ⊕ Cofactors, Vitamins, Prosthetic Groups, Pigments (140)
- ⊕ Cell Wall and Capsule (17)
- ⊕ Virulence, Disease and Defense (33)
- ⊕ Potassium metabolism (9)
- ⊕ Photosynthesis (0)
- ⊕ Miscellaneous (43)
- ⊕ Phages, Prophages, Transposable elements, Plasmids (3)
- ⊕ Membrane Transport (86)
- ⊕ Iron acquisition and metabolism (23)
- ⊕ RNA Metabolism (50)
- ⊕ Nucleosides and Nucleotides (71)
- ⊕ Protein Metabolism (187)
- ⊕ Cell Division and Cell Cycle (2)
- ⊕ Motility and Chemotaxis (0)
- ⊕ Regulation and Cell signaling (32)
- ⊕ Secondary Metabolism (23)
- ⊕ DNA Metabolism (56)
- ⊕ Fatty Acids, Lipids, and Isoprenoids (78)
- ⊕ Nitrogen Metabolism (9)
- ⊕ Dormancy and Sporulation (1)
- ⊕ Respiration (82)
- ⊕ Stress Response (64)
- ⊕ Metabolism of Aromatic Compounds (98)
- ⊕ Amino Acids and Derivatives (281)
- ⊕ Sulfur Metabolism (48)
- ⊕ Phosphorus Metabolism (27)
- ⊕ Carbohydrates (177)

Figure 3

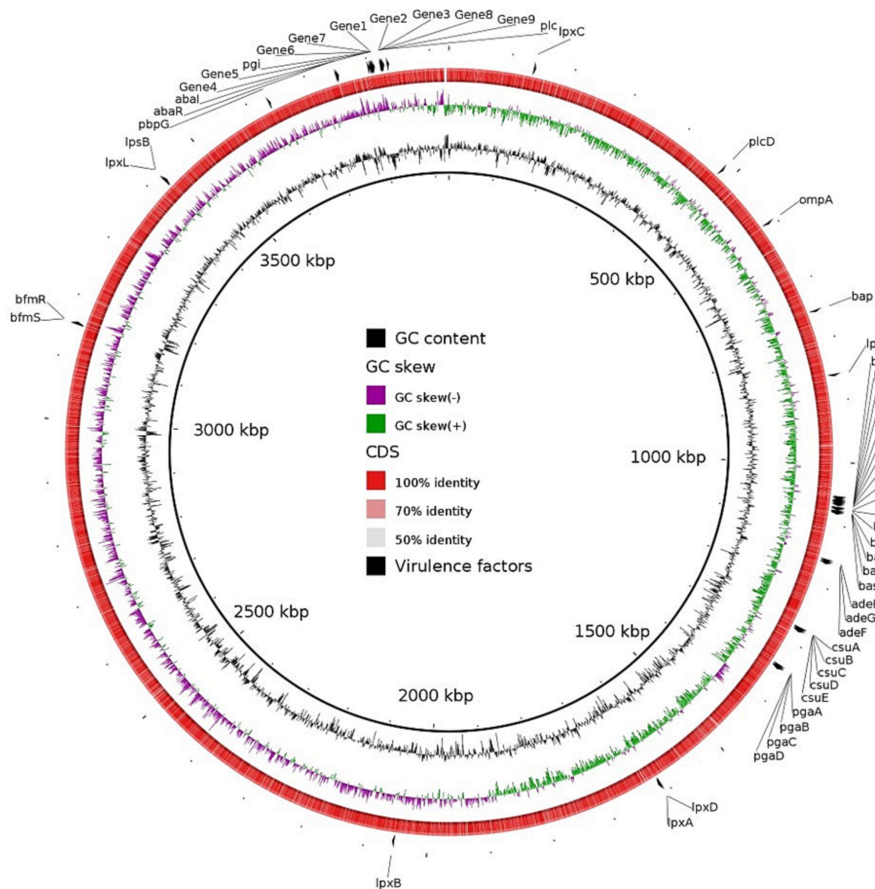


Figure 4

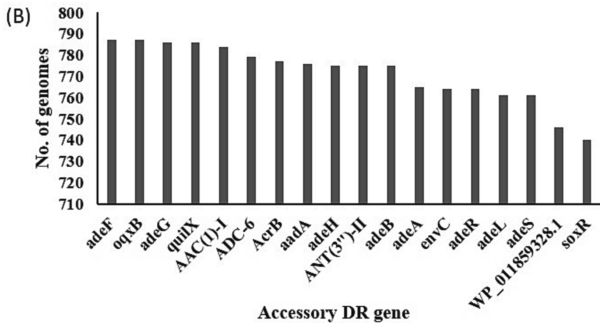
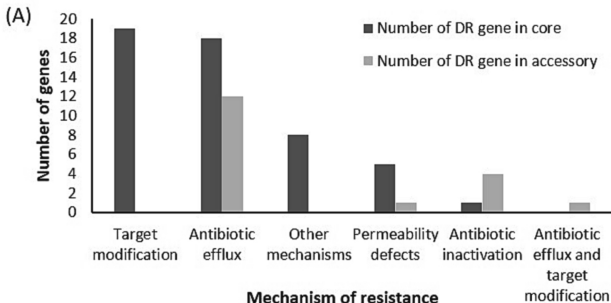


Figure 5